Techfest 2026-27

The Wrecking Ball Challenge

Round 1: Submission Template

TEAM IDENTIFICATION

- **Team ID & Name:** TM-CB1C8E0CEAA | EigenThinker (robot: RoToRBot)
- **Institution:** IIIT-Delhi (Indraprastha Institute of Information Technology Delhi)
- **Team Lead Contact:** Sameer Saawan | sameer.saawan10@gmail.com | 9289528242

Design signature: RoToRBot

- **Twin-Eye axle:** two cameras on the ends of a non-rotating axle, held level by an internal servo. Each camera handles one cone colour (left yellow, right red), and their overlap gives stereo range on a 0.73 m baseline.
- **Ballast-sled steering:** the battery and steel ballast are the steering mass. One stepper slides 4.9 kg ±70 mm, so the robot carries no extra weight just to steer.
- **Torque-vector control:** carried over from our earlier spherical robot. The controller commands a pitch-and-roll torque vector and maps it linearly onto independent H-bridge and stepper channels.
- **Gate-and-Cue state machine:** a sector changes only when a distance window and a sensor cue agree, so shell slip cannot mis-sequence the run.
- **Radio discipline:** BLE for bench tuning and Wi-Fi for logs before and after a run, both switched off during the run; only the kill receiver stays live.
- **INNOVATION (future):** a Kandó-inspired adaptive capacitor network, our own power concept,described in section 2.

1. Mechanical Architecture & Enclosure Design (CAD Evidence)

- **CAD Renders Provided:** the full assembly is a CAD model of 83 named parts (STEP file available), with every component modelled to its real dimensions.
    - Isometric views, external and internal (shell shown transparent): Figures 1 and 2.

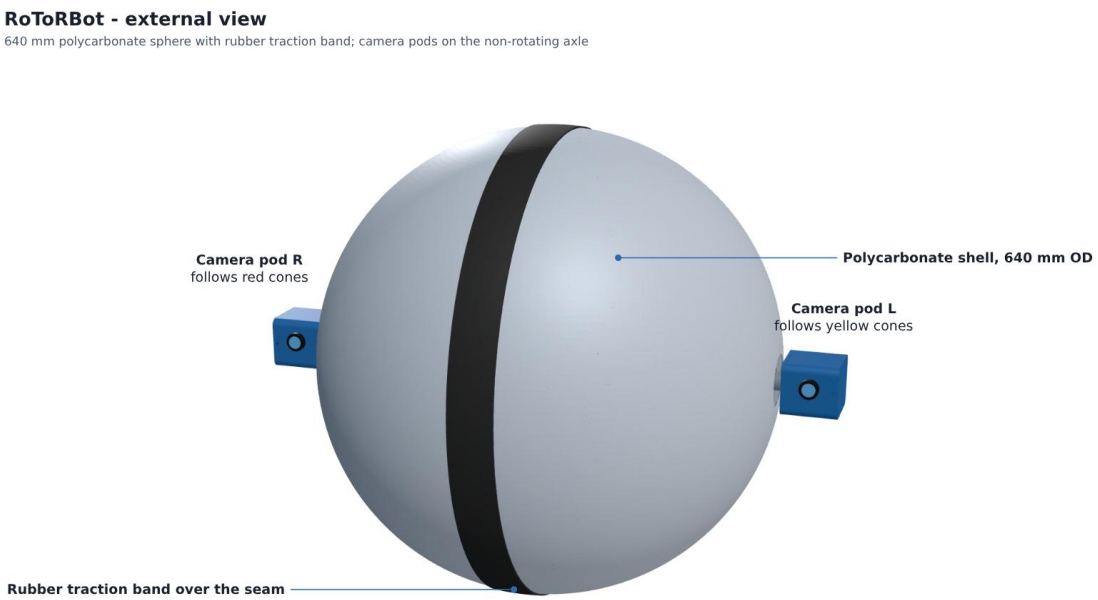


Figure 1. Isometric view, external: 640 mm polycarbonate sphere, traction band, camera pods on the fixed axle.

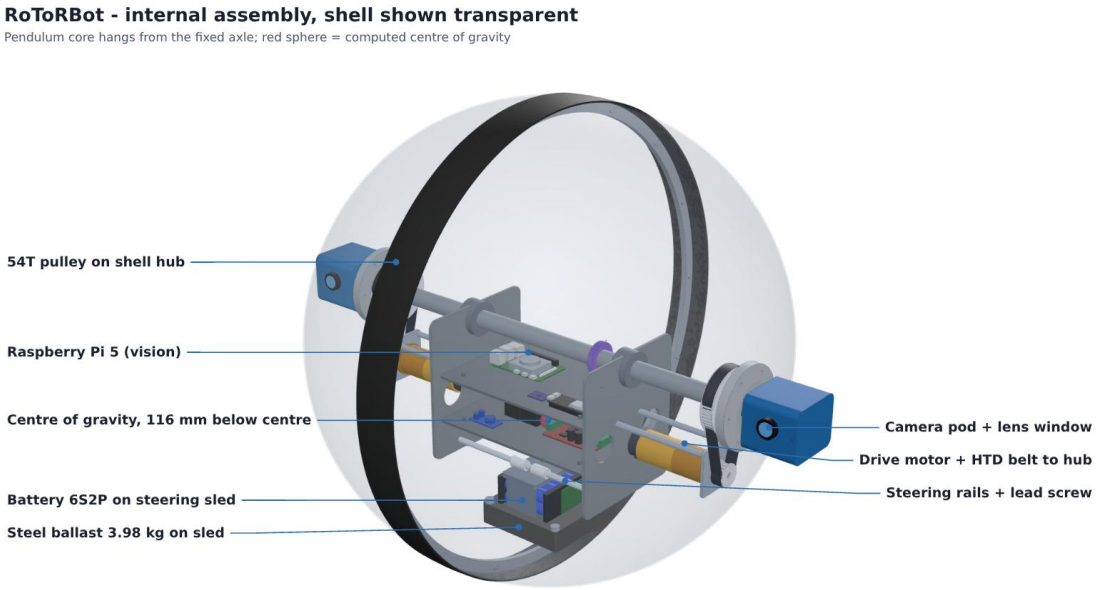


Figure 2. Isometric view, internal (shell transparent): pendulum core, drive, sled; red sphere = computed CoG.

- Cross-sections at the rolling plane (dimensions and computed CoG) and through the drive train: Figures 3 and 4.

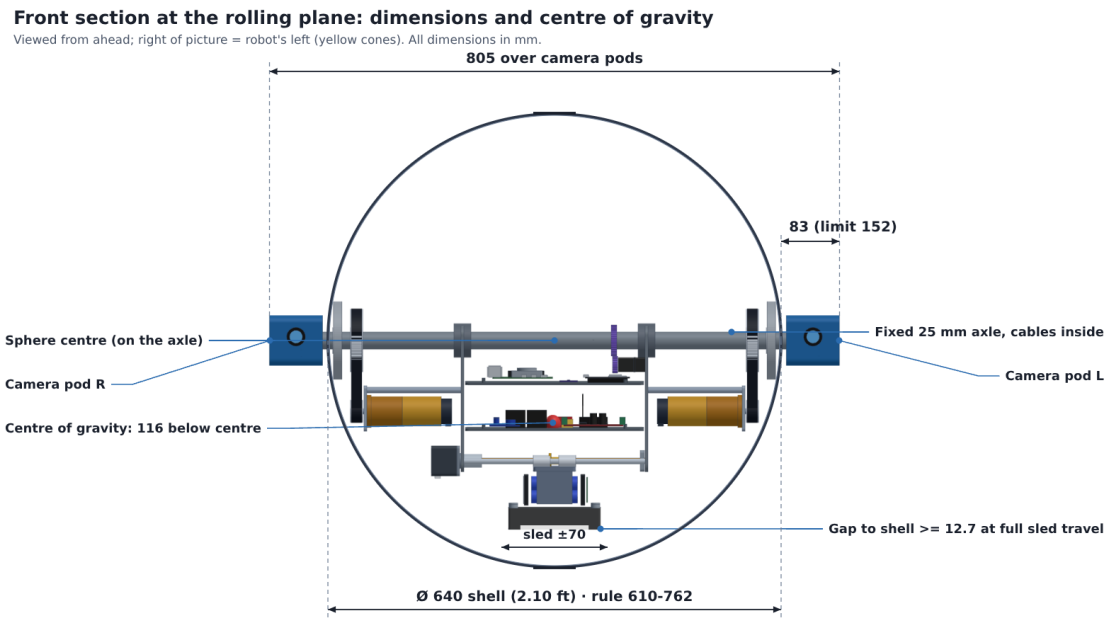


Figure 3. Cross-section at the rolling plane: diameter, pod protrusion, sled travel, clearance and CoG (mm).

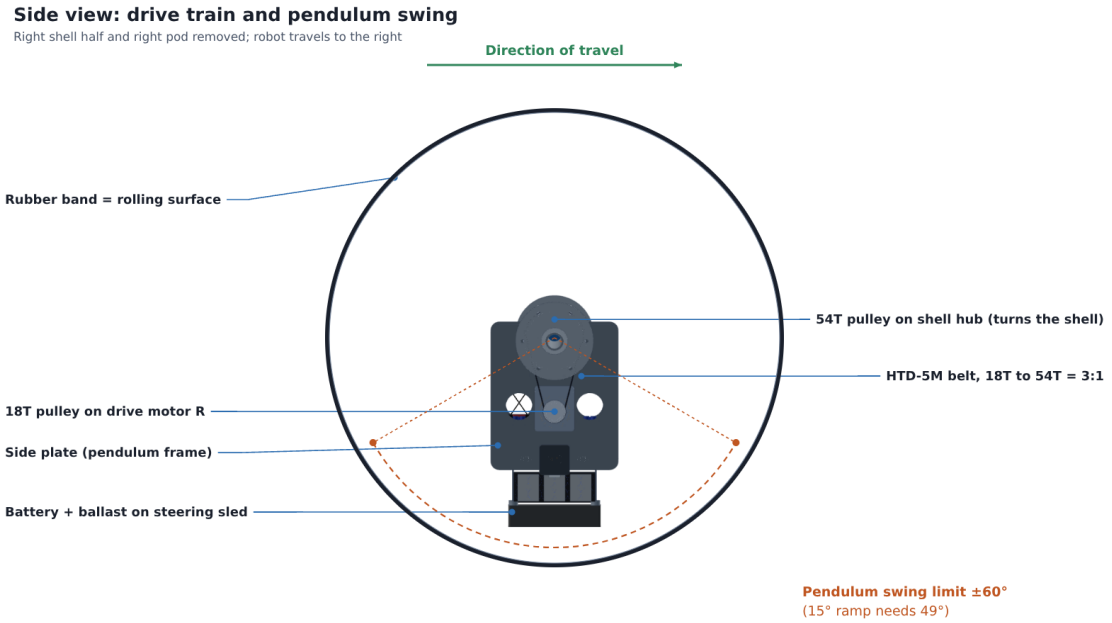


Figure 4. Cross-section through the drive train: 18T to 54T belt drive and the ±60° pendulum swing limit.

- **Shell Integrity:**
  - *Material:* Thermoformed clear polycarbonate: two hemispheres from 3 mm sheet (about 2 mm wall after forming), bolted at an internal equator flange so the outside stays flush. A 60 mm rubber traction band covers the seam.

    - *Exact Outer Diameter:* 640 mm = 2.10 ft (644 mm = 2.11 ft over the traction band), inside the 2.0-2.5 ft rule with 30 mm of margin on the minimum.
- **The 0.5-Foot Limit Rule:** Confirmed. The only external parts are the two camera pods on the non-rotating ends of the axle. Each pod ends 82.5 mm (0.27 ft) beyond the outer shell surface, against the 152.4 mm (0.5 ft) limit, leaving 70 mm spare; overall width over the pods is 805 mm (Figure 3). No other mount, lens or part protrudes.
- **CoG & Drive Mechanism:** Pendulum drive with ballast-sled steering. Two 24 V gearmotors hang on an internal pendulum and turn the shell through 18T to 54T HTD-5M belts (3:1). The motor reaction swings the pendulum forward, and its offset weight rolls the ball; a PD loop on the pendulum angle damps the rocking. To steer, a NEMA 17 stepper and T8 lead screw slide the 4.91 kg battery-and-ballast sled ±70 mm along the axle, tilting the ball up to 12° so it curves (about 1.5 m radius). A servo keeps the camera axle level. Calculated CoG relative to the geometric centre (X forward, Y left along the axle, Z up): X = -0.2 mm, Y = -2.4 mm (trimmed to 0 by a +6.8 mm sled offset), Z = -116.2 mm. Total mass 14.0 kg; pendulum moment Σ ml = 1.625 kg·m; restoring moment 2.8 N·m at 10 degrees of tilt.

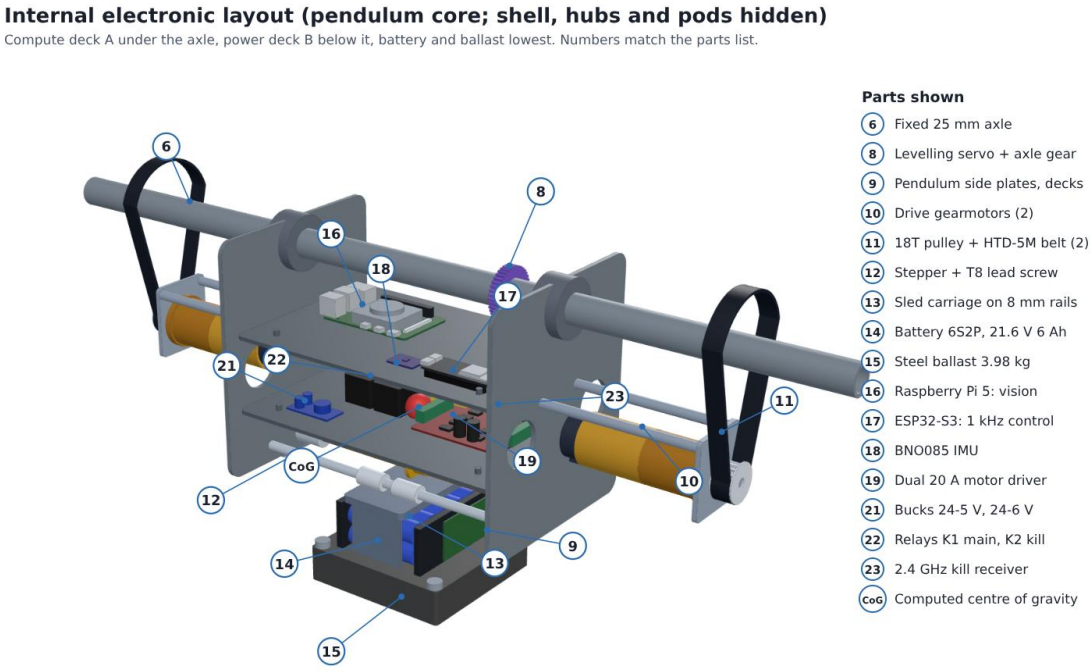


Figure 5. Internal layout of the pendulum core, modelled to real component dimensions.

- **Ramp-Climbing Capability:** The 0.8 ft (244 mm) ramp sizes the drive. Steady-climb shell torque is $\tau = m \cdot g \cdot R \cdot (\sin\theta + C_{rr} \cdot \cos\theta)$, with m = 14.0 kg, R = 0.32 m and C_{rr} = 0.015 (table below). The design peak is 17.5 N·m (17° plus 30 % for acceleration); through two motors and 3:1 belts at 95 %

efficiency, each motor needs 3.1 N·m at 134 rpm (1.5 m/s). Selected: 24 V, 45 mm planetary gearmotors rated ≥ 3.1 N·m continuous at 140 rpm, about 190 rpm no-load, stall ≥ 6.5 N·m. The pendulum, not the motor, limits a steady climb to 18.3° at the 60° swing limit. If the 0.8 ft ramp is steeper, a 2.0 m/s run-up stores 33.6 J, enough to lift the ball 245 mm (the full 0.8 ft) before the drive adds anything. Grip needed at 15°: μ ≥ 0.27, met by the rubber band.

Ramp angle	Pendulum swing needed	Shell torque	Torque per motor
10°	31°	8.3 Nm	1.5 Nm
15°	49°	12.0 Nm	2.1 Nm
17°	58°	13.5 Nm	2.4 Nm
18.3° (steady-climb limit)	60°	14.4 Nm	2.5 Nm

2. Electrical & Power System Architecture

- **Electrical Block Diagram:** Figure 6 (drawing ET-RRB-E01) shows power distribution, the two controllers, every sensor interface and both motor-driver paths, with the ESP32-S3 pin number on each connection. Radio discipline: BLE tuning and Wi-Fi logs are switched off during the run, so only the kill receiver stays live.

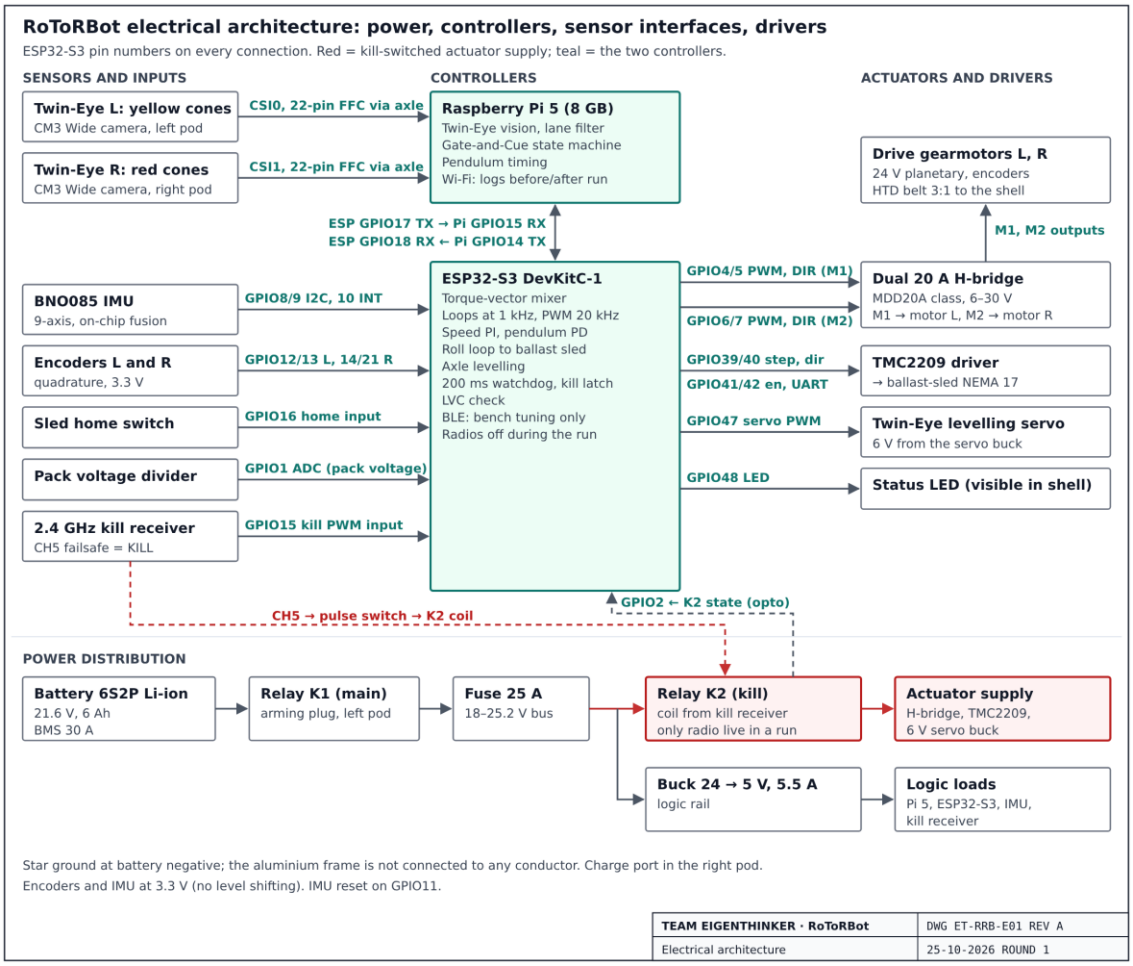


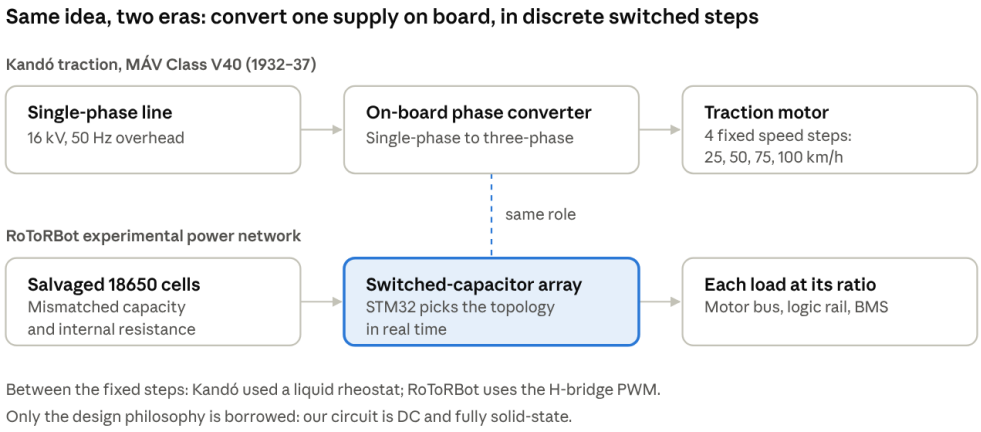
Figure 6. Electrical architecture (ET-RRB-E01): ESP32-S3 pin assignments and the kill-switched actuator supply.

- **Power Source & BMS:**
  - *Confirmation:* Strictly battery-powered: Yes. No tether, cable or external power feed.
  - *Chemistry, Voltage & Capacity:* Li-ion, 18650 high-drain cells (≥ 15 A each) in 6S2P: 21.6 V nominal (25.2 V full, 18.0 V empty), 6 Ah, 130 Wh.
  - *Safeguards:* 6S BMS rated 30 A (over-charge, over-discharge and short-circuit protection, cell balancing) and a 25 A main fuse. A software low-voltage cut-off on the ESP32 reads the pack through an ADC divider: warning at 19.8 V (3.3 V/cell), controlled stop at 19.2 V (3.2 V/cell) under load. The BMS hard cut-off is the last layer.
  - **INNOVATION - Experimental: Kandó-inspired adaptive capacitor network (our long-term goal)**

    Our own power concept, first proposed for our earlier spherical robot, replaces fixed converters with a switched-capacitor network that an

STM32 reconfigures in real time. It borrows only the design philosophy of Kálmán Kandó's railway traction system.

Kandó's MÁV Class V40 locomotives (1932 – 37 source : Wiki) took a single-phase 16 kV, 50 Hz line and converted it on board to three-phase. The motor ran at four fixed speeds, with a liquid rheostat for the speeds in between. Our network does the same job for a DC robot: one supply, converted on board, in discrete switched steps.



Kandó traction and the RoToRBot power network , V40 facts from Wikipedia

- **Switched-capacitor array:** MOSFETs switch banks of capacitors between series and parallel arrangements. Each arrangement transfers energy at a fixed voltage ratio, acting like a transformer with no magnetic core.
- **STM32 controller:** it reads every cell voltage, the bus voltages and the load currents, and changes the topology in real time.
- **What it balances:** energy flow between the sensor and communication loads, the PWM motor drivers, the BMS, and the uneven discharge curves of low-grade cells salvaged from broken electronics.
- **Safety boundary:** the network sits after relay K1 and before relay K2, so the kill path stays hardware-only. A bypass relay returns the robot to the baseline system at any time.
- **Status:** bench prototype. The competition robot runs on the baseline system above, and the module is fitted only after it passes bench tests. Its estimated mass, under 0.25 kg, fits inside the 0.30 kg contingency.

Improvements this power system targets

The adaptive network is aimed at seven improvements over a fixed battery-plus-converter system. Each has a target we will measure on the bench and report in Phase 2.

Improvement	How the network delivers it	Target to measure
Steady motor-bus voltage	Steps the capacitor ratio up as the pack sags, so torque per PWM step stays constant and the sizing model holds for the whole run	Bus within ±5 % from full to empty; the bare pack falls 29 % (25.2 to 18.0 V)
Lower battery peak current	Charged capacitors supply the short bursts of acceleration and ramp climbing	Peak battery current 30 % lower on the 0.8 ft ramp
More energy from salvaged cells	Moves charge from strong cells to weak ones instead of bleeding the strong ones as heat	15-25 % more usable capacity than passive balancing
No logic brownouts	The logic rail is fed from its own capacitor stage, isolated from motor dips	No Pi or ESP32 reset in a stalled-motor test
Braking energy recovered	Braking before the pendulum and on ramp descents charges the array	Store the 19 J of one stop from 1.5 m/s
Lighter, quieter conversion	No inductors; switching at fixed ratios	Less magnetic noise near the IMU and radios
Battery health data	The STM32 logs every cell and the ratio it chose	Per-cell discharge curves for choosing cells

- **Actuation & Motor Control: (MAY CHANGE AS PER COST EFFICIENCY)**
    - *Motor Specs:* Drive (×2): 24 V, 45 mm planetary gearmotor (775-size motor) with quadrature encoder; required ≥ 3.1 Nm continuous at 140 rpm, about 190 rpm no-load, stall torque ≥ 6.5 Nm (≈ 66 kgf·cm). Steering: NEMA 17 stepper (42 × 40 mm, about 0.4 Nm holding) on a T8 lead screw. Camera levelling: metal-gear digital servo, about 20 kgf cm.
    - *Driver Boards:* Propulsion: dual-channel brushed DC H-bridge (Cytron MDD20A class), 6–30 V, 20 A continuous per channel, PWM up to 20 kHz, with a firmware current limit of 10 A per motor. Steering: TMC2209 stepper driver (up to 2 A RMS, configured over UART, silent StealthChop mode). Levelling servo: PWM from the ESP32, powered by a 24 to 6 V, 3 A buck converter.

- **Safety Fail-Safes:**
    - *Wireless Kill-Switch:* A separate 2.4 GHz RC transmitter and receiver carry only the kill channel, so the run stays fully autonomous. The receiver drives relay K2, which feeds both motor-driver channels, the stepper driver and the servo supply. K2 stays closed only while valid RUN pulses arrive: a thrown switch, a transmitter switched off, a lost signal or an unpowered receiver all open it, and the receiver failsafe is also set to KILL. The ESP32 reads the channel and K2's state too, zeroes every PWM output and latches a fault, so nothing restarts until the robot is re-armed. We will demonstrate each path with the shell spinning on a stand at Day 1 inspection.
    - *Manual Disconnect:* A keyed arming plug (XT60 loop) in the left camera pod, on the non-rotating end of the axle at 320 mm height, reachable from outside at any time, even while the shell rolls. Pulling it opens main relay K1 and removes all power. The charge port is in the right pod.

3. Software & Vision-Based Navigation Pipeline

- **Software Flowchart:** Figure 7 (drawing ET-RRB-S01) shows the Gate-and-Cue run state machine (left) and the Twin-Eye vision stack from camera frame to the torque-vector mixer (right).

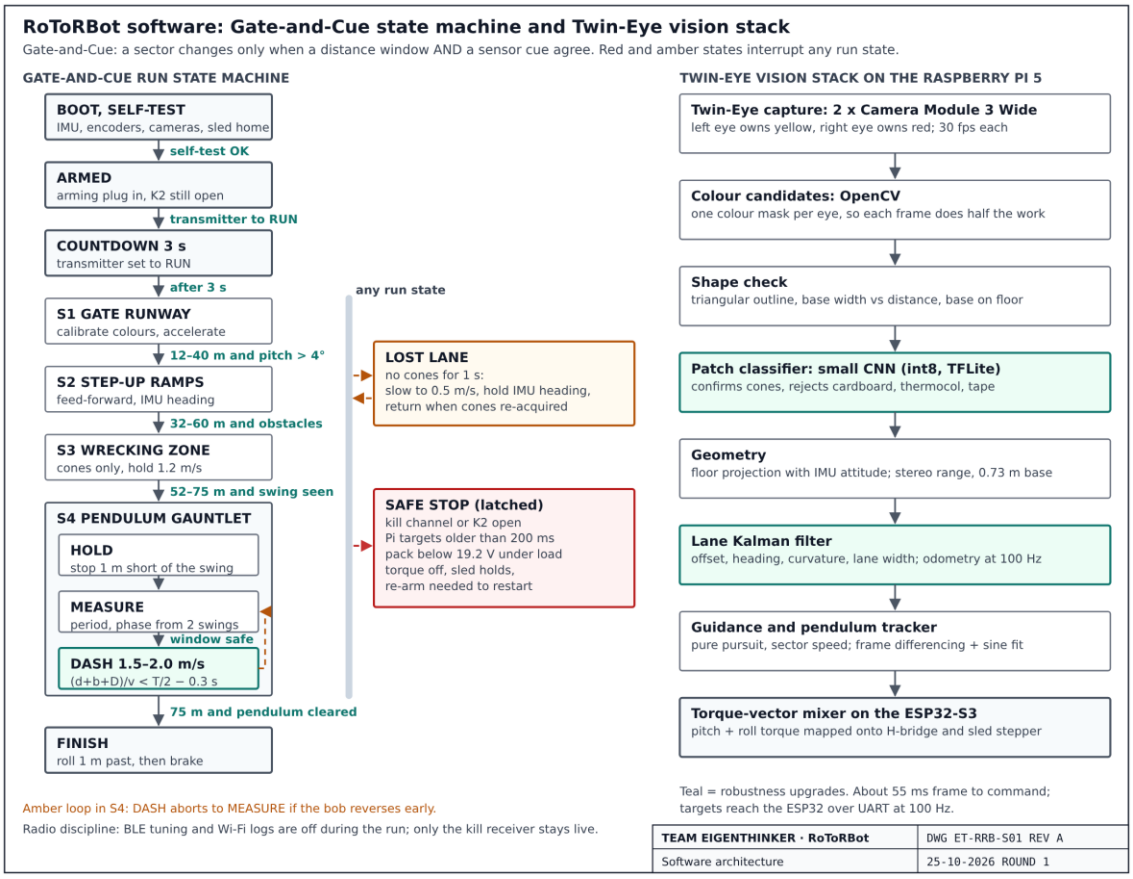


Figure 7. Software architecture (ET-RRB-S01): Gate-and-Cue state machine and Twin-Eye vision stack.

- **Primary Vision Processing:** Twin-Eye vision: two Raspberry Pi Camera Module 3 Wide cameras, one in each pod (0.73 m stereo baseline, kept level by the axle servo), stream over CSI to a Raspberry Pi 5 at 30 fps (768 × 432). Each eye owns one cone colour, left yellow and right red (two hue bands), so every frame runs a single OpenCV HSV mask and does half the work. Each candidate must pass a shape check (triangular outline, base width matching its distance, base on the floor) and a small int8 CNN patch classifier trained on our practice-day images, which rejects cardboard, thermocol or tape of a similar colour. Exposure and white balance are locked after calibrating on the Sector 1 chevron gates. Each cone base is projected onto the floor from the 320 mm camera height and the IMU attitude, and cones seen by both cameras also get stereo range.
- **State Estimation & Control:** The BNO085 IMU fuses gyro and accelerometer on-chip (game rotation vector, no magnetometer near the motors) for pitch, roll and yaw. Ground speed combines the motor encoders with the pendulum's pitch rate, $v = R(\omega_{rel} + d\varphi/dt)$. A Kalman filter tracks lane offset, heading error, curvature and lane width: odometry predicts it at 100 Hz, each cone corrects it, outlier cones are rejected, and if one side's cones are hidden it follows the visible side using the learned lane width. On

the ESP32 at 1 kHz, a speed PI loop asks for a pendulum angle and a PD loop on that angle damps the 1.6 s rocking mode; a roll loop sets the ballast sled to steer, and the torque-vector mixer maps pitch and roll torque onto the H-bridge and the stepper. The Gate-and-Cue state machine changes sector only when a distance window and a sensor cue agree, so wheel slip cannot mis-sequence the run.

- **Obstacle Handling Logic (Sector 4):** The Pendulum state starts only when odometry reads 52–75 m and a swinging object is detected ahead. RoToRBot stops 1 m short; both cameras follow the bob by frame differencing in a window above the track, and stereo gives its position across the lane. A sine fit over two swings gives the period T and the phase. It dashes at 1.5–2.0 m/s only if $(d + b + D) / v < T/2 - 0.3$ s ($d = 1$ m stand-off, b = bob depth, $D = 0.64$ m ball, v = dash speed), starting as the bob clears the lane. If the fitted period varies by more than 10 % or the bob reverses early, it brakes and waits for the next swing.

Supporting files: detailed Phase 1 CAD & System Document (full calculations, mass budget, bill of materials, risks) and the 83-part STEP assembly: